

Complete rational subgroup preimages

Hidden division relations and exact witness saturation

All coefficient lines over F2 and F3, complete rational prime fibres, exact replacement generators and a witness subgroup saturated at both primes.

10

worked prime-preimage packets

40

maximum projective coefficient
lines

2 / 3

hidden relative indices
recovered

Complete rational preimage generators for supplied subgroups with at most four sources. Prime saturation of a specified witness subgroup; ambient rank, other primes and global integral bounds remain open.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

The new operation computes generators for $D_p(G) = \{H \in E(Q) : pH \text{ lies in } G\}$, where G is generated by up to four supplied rational points and p is 2 or 3. Its mathematical conclusion covers every rational H and every integer combination of the supplied generators. The finite computation enumerates residue coefficient lines; it does not bound the coordinates of H or the integer coefficients defining G . Exact complete doubling or tripling fibres decide each residue class. Reduced row echelon relations then give replacement generators, and the full rational p -torsion kernel is retained. Dependent points, infinity and torsion are supported without an independence premise.

The worked two-generator examples expose divisibility missed by testing either generator separately. They recover the rational witness subgroup generated by $P=(0,1)$ and $Q=(2,1)$ on $y^2=x^3-4x+1$ from two different sublattices. The first has index two, generated by $P+Q$ and $P-Q$; the second has index three, generated by $2P+Q$ and $P+2Q$. The recovered witness subgroup is closed under rational division by both two and three. Consequently it is closed under every multiplier $2^a 3^b$, for arbitrary nonnegative exponents. This is saturation of a specified subgroup at specified primes; it does not identify the entire Mordell-Weil group or settle other primes.

Integration and scope

The consolidated base for this release is `fc89cf403b33a6a029c81a0ed865ceaaf2f6c9c6`. Incoming work since the composite-division release adds native rational-root and torsion emitters, residue-population proofs, marked Picard-Lefschetz braid actions and exact polynomial flavor trial bounds. The commits and their receipts are preserved. This extension uses the complete rational [2] and [3] fibres already supplied by the elliptic arithmetic interface, together with their existing Python replay. The new production modules are `elliptic_subgroups.py` and `elliptic_subgroup_verifier.py`, wired into `elliptic_arithmetic.py` and `query_service.py`.

The output is a complete finite generating set for a generally infinite subgroup. It is neither an enumeration of every group element nor a claimed complete rank calculation for the ambient curve. No global height bound, integral-point census or rank upper bound is needed for this inverse-subgroup calculation. The point laws, root search and interpreter retain their existing Python trust boundary. The latest Lean bridges establish their stated native instances and group foundations; this release adds no new Lean theorem.

The finite relation space

Let $P_1, \dots, P_r$ generate G over the integers. Consider the linear map from $\mathbb{F}_p^r$ to $E(Q)/pE(Q)$ sending a coefficient vector v to the class of the sum $\sum v_i P_i$. This is well-defined: changing an integer lift by p times another vector changes the point by an element of $pE(Q)$. Its kernel R consists exactly of residue combinations having a rational p -preimage. Thus R is a finite vector subspace, even when the supplied points are dependent or torsion.

Every nonzero coefficient vector lies on exactly one projective line represented by zeros before its first nonzero coordinate, followed by 1 and an arbitrary tail. There are $(p^r-1)/(p-1)$ such lines: at most fifteen for doubling and forty for tripling with four source generators. Membership is constant on a line. If c is nonzero modulo p and $cS=pH$, choose integers a, b with $ac+bp=1$; then $S=p(aH+bS)$. The reverse implication follows by multiplying an existing preimage by c .

The producer tests every normalized line with a complete rational division packet for its literal coefficient sum. The zero vector has fibre $E(Q)[p]$, computed separately and completely. A missing rational square lift or signed point equality is a genuine restriction; it must not be discarded when reducing an x -coordinate equation. The existing prime-fibre packets retain those restrictions. A bounded anchor-discovery failure is not evidence of emptiness: an empty prime fibre must retain the exhaustive rational-root evidence required by the previous release.

The producer row-reduces all divisible line representatives over F_p . Replay independently constructs the projective domain, checks every complete fibre, verifies the proposed row echelon form, enumerates its entire span, and compares that span with every scalar multiple of every positively classified line. It does not reuse the producer's elimination algorithm or root discovery. An omitted line, changed coefficient sum, fabricated relation or incomplete basis is rejected.

Complete generators without an independence assumption

Choose the reduced row echelon basis $b_1, \dots, b_d$ of R and rational anchors $H_1, \dots, H_d$ satisfying $pH_j = \sum_i b_{ji} P_i$, using literal coefficient representatives from 0 through $p-1$. Each row begins with a pivot coefficient exactly 1; all other pivot columns are zero. Therefore

$$P_{\text{pivot}(j)} = p H_j - \sum_{\text{nonpivot } i} b_{ji} P_i.$$

Replace each pivot source generator by its anchor, leave the nonpivot source generators in place, and adjoin the complete rational p -torsion kernel K . Let G_{new} be the generated subgroup. The displayed identities imply that G_{new} contains every original P_i . It also contains every anchor and every kernel point. Conversely p times every replacement lies in G , and $pK=0$, so G_{new} is contained in $D_p(G)$.

To prove the other inclusion, take H with $pH = \sum_i n_i P_i$ for integers n_i . Write $n_i = pm_i + v_i$ with every component of v in $\{0, \dots, p-1\}$. Then v lies in R . Express v modulo p as a combination $\sum_j c_j b_j$ and write $v - \sum_j c_j b_j = pt$ for an integer vector t . It follows that

$$p (H - \sum_j c_j H_j - \sum_i (m_i + t_i) P_i) = 0.$$

The expression in parentheses lies in K . Since G_{new} contains the anchors, all original source points and K , it contains H . Hence $G_{\text{new}} = D_p(G)$. This proves completeness over all rational points and unbounded integer source coefficients using only the finite residue computation.

The proof makes no assertion that the source presentation is injective. If two source generators are equal, a nonzero coefficient relation can occur even though the subgroup is already p -saturated. Four identity source points give relation dimension four while generating the zero subgroup. Therefore relation dimension alone is not an index exponent for arbitrary input. Any index claim requires the relevant independence or lattice-presentation hypotheses. The corpus retains dependent and zero examples specifically to prevent that misinterpretation.

Hidden index-two and index-three examples

On $y^2=x^3-4x+1$, $P=(0,1)$ and $Q=(2,1)$ have an independently replayed witness-span lower bound two. Thus they form a free basis of their generated subgroup, without any claim that the ambient curve has rank exactly two.

For $A=P+Q$ and $B=P-Q$, neither A nor B has a rational half. Their sum is $2P$, however. The three projective coefficient tests over F_2 find precisely the nonzero relation $(1,1)$. The replacement list is P,B . Since $Q=P-B$, this list generates the original P,Q witness subgroup. The source lattice has determinant -2 relative to P,Q , and the replacement has determinant -1 . This gives an exact relative index-two enlargement.

For $A=2P+Q$ and $B=P+2Q$, neither A nor B has a rational third. Their sum is $3(P+Q)$. The four projective tests over F_3 again find the relation $(1,1)$. The replacement list is $H=P+Q,B$. Now $P=2H-B$ and $Q=B-H$. The source coefficient determinant is 3 and the replacement determinant is 1. This gives the exact relative index-three enlargement. The anchors are computed from complete fibres rather than supplied as hoped-for divisions.

The follow-up packets for source points P,Q have no nonzero division relations for either prime, and their rational kernels contain only infinity. Their output generators are exactly P,Q . By the complete-preimage theorem, $D_2(G)=G$ and $D_3(G)=G$. Repeated inverse multiplication therefore gives $D_{(2^a 3^b)}(G)=G$ for every nonnegative a,b , with no fixed upper bound on the multiplier. A further prime or an ambient group generator could still enlarge the mathematical picture.

Torsion, dependencies and composition

On $y^2-3xy-12y=x^3-12x^2$, the point $U=(0,0)$ has order nine and $T=3U$ has order three. Applying the prime-preimage operation to the subgroup generated by T produces generators for the full order-nine subgroup. Retaining only the anchor would omit the possible p -kernel ambiguity; the implementation retains the complete kernel. The corpus checks the resulting finite generated set against all nine multiples of U .

Duplicate generators on the rank-one Mordell model show a nonzero coefficient relation that does not enlarge the subgroup. Identity source points show the extreme presentation dependence. The generalized rational model with coefficients $(1, -1/4, 1, -1/2, -9/4)$ recovers $(3,3)$ from its triple inside the subgroup calculation, retaining original y -coordinate conventions.

On $y^2=x^3-2$, start with G generated by $6(3,5)$. Its complete doubling preimage is generated by $3(3,5)$, and the complete tripling preimage of that subgroup is generated by $(3,5)$. Each stage is a full subgroup statement. In general $D_p(D_q(G))=D_{(pq)}(G)$, because pH lies in $D_q(G)$ exactly when qpH lies in G . Users can compose the operation while each returned generating set remains within the four-source input scope. Redundant kernel generators can exceed that input bound; an automatically reduced presentation and a broader iterative saturation driver remain next work. No partial completeness claim is returned when the next request exceeds its contract.

Public execution and resource bounds

A persisted `elliptic_curve` now exposes `subgroup_preimage(points,prime,node_limit)`. The prime is exactly 2 or 3, with at most four canonical rational source points; infinity is JSON null. The response retains the original model and source list, complete kernel fibre, every normalized projective fibre, canonical relation basis, replacement equations, replacement points, complete generated point list and request-wide work accounting. It explicitly declines a complete ambient Mordell-Weil group claim.

One root-node budget, between 1 and 100000, covers the kernel and all residue fibres. Each subsequent packet records the remaining budget. There is no per-line renewal. Coefficient and nested-packet allocation limits inherited from the complete prime division implementation apply as well. A large curve model or source combination may exceed these limits; the operation then fails explicitly. The width-four bound caps the combinatorial domain at forty lines and avoids an uncontrolled exponential scan.

Run `make elliptic-subgroups` to execute the focused tests and regenerate the corpus. The generator is `python/develop_elliptic_subgroups.py`. Ten preimage packets, one witness-independence packet and four cold-reload service responses are bound by `receipts/elliptic_subgroups/summary.json`. The validation document retains exact test scopes and transcript locations. The standalone service operation `verify_elliptic_subgroup_preimage` performs bounded replay of a packet without discovering roots.

The next mathematical bridge

The complete prime-preimage operation is a foundation for p -saturation algorithms: it tests all coefficient combinations, constructs actual rational enlargement generators and certifies closure when the generated subgroup is unchanged. The next arithmetic work is a reduced integer presentation that controls redundant generators and torsion through repeated extensions, followed by primes beyond three. Termination and a complete ambient basis require additional Mordell-Weil information, such as rank and index bounds; repeatedly applying the current operation alone does not establish them.

The next formal step is a generic subgroup-preimage generation theorem, followed by native finite-field semantics and point-law refinement. Its generation statement must cover dependent source presentations; relative index statements need their separate independence hypotheses.